

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Sunday 1 March 2026

1.1 Daily Limit

Limit of a Non-linear Recursive Sequence

Evaluate the limit:

$$\lim_{n \rightarrow \infty} a_n \text{ where } a_0 = 3, a_1 = 2, \text{ and } a_n = \sqrt{a_{n-1}a_{n-2}} \text{ for } n \geq 2$$

Solution: Taking the natural logarithm of both sides of the recurrence relation, we obtain a linear recurrence:

$$\ln(a_n) = \frac{1}{2} \ln(a_{n-1}) + \frac{1}{2} \ln(a_{n-2})$$

Let $b_n = \ln(a_n)$. The sequence becomes $b_n = \frac{1}{2}b_{n-1} + \frac{1}{2}b_{n-2}$ with initial conditions $b_0 = \ln 3$ and $b_1 = \ln 2$. The characteristic equation is:

$$r^2 - \frac{1}{2}r - \frac{1}{2} = 0 \implies 2r^2 - r - 1 = 0 \implies (2r + 1)(r - 1) = 0$$

The roots are $r_1 = 1$ and $r_2 = -1/2$. Thus, the general solution for b_n is:

$$b_n = c_1(1)^n + c_2\left(-\frac{1}{2}\right)^n$$

As $n \rightarrow \infty$, the term $(-1/2)^n \rightarrow 0$, so $\lim_{n \rightarrow \infty} b_n = c_1$. Using the initial conditions to solve for c_1 :

$$c_1 + c_2 = \ln 3$$

$$c_1 - \frac{1}{2}c_2 = \ln 2$$

Multiplying the second equation by 2 gives $2c_1 - c_2 = 2 \ln 2$. Adding this to the first equation yields:

$$3c_1 = 2 \ln 2 + \ln 3 = \ln(4) + \ln(3) = \ln(12)$$

Thus, $c_1 = \frac{1}{3} \ln(12) = \ln(12^{1/3})$. Since $b_n \rightarrow \ln(\sqrt[3]{12})$, we exponentiate to find the limit of the original sequence a_n :

$$\lim_{n \rightarrow \infty} a_n = e^{\ln(\sqrt[3]{12})} = \sqrt[3]{12}$$

Monday 2 March 2026

2.1 Daily Limit

Limit of Nested Radicals and Inverse Sine

Evaluate the limit:

$$\lim_{x \rightarrow 0^+} \frac{\sqrt{1 - \sqrt{1 - x^2}}}{\sin^{-1}(x)}$$

Solution: To simplify the denominator, let us substitute $x = \sin(\theta)$. As $x \rightarrow 0^+$, we have $\theta \rightarrow 0^+$. Thus, $\sin^{-1}(x) = \theta$. Substituting this into the numerator:

$$\sqrt{1 - \sqrt{1 - \sin^2(\theta)}} = \sqrt{1 - \cos(\theta)}$$

We can use the half-angle identity, $1 - \cos(\theta) = 2 \sin^2\left(\frac{\theta}{2}\right)$. The expression becomes:

$$\sqrt{2 \sin^2\left(\frac{\theta}{2}\right)} = \sqrt{2} \left| \sin\left(\frac{\theta}{2}\right) \right|$$

Since $\theta \rightarrow 0^+$, $\sin\left(\frac{\theta}{2}\right)$ is positive, allowing us to drop the absolute value. The limit translates to:

$$L = \lim_{\theta \rightarrow 0^+} \frac{\sqrt{2} \sin\left(\frac{\theta}{2}\right)}{\theta}$$

We can multiply the numerator and denominator by $\frac{1}{2}$ to use the standard limit $\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1$:

$$L = \sqrt{2} \lim_{\theta \rightarrow 0^+} \frac{\sin\left(\frac{\theta}{2}\right)}{2 \cdot \left(\frac{\theta}{2}\right)} = \frac{\sqrt{2}}{2} (1) = \frac{1}{\sqrt{2}}$$

Tuesday 3 March 2026

3.1 Daily Limit

Double Limit of a Parameterized Function

Consider the sequence L_k for $k \in \mathbb{Z}^+$ defined by:

$$L_k = \lim_{x \rightarrow k} \left[\left(\frac{\sin(7\pi x)}{\sin(\pi x)} \right)^{\frac{x}{x+1}} \right]$$

Evaluate $\lim_{k \rightarrow \infty} L_k$.

Solution: First, we evaluate the inner limit L_k . Since $x \rightarrow k$ (where k is an integer), both the numerator and denominator of the fraction $\frac{\sin(7\pi x)}{\sin(\pi x)}$ approach 0. We can apply L'Hôpital's Rule or use a substitution $x = k + h$ where $h \rightarrow 0$. Using substitution:

$$\sin(7\pi(k + h)) = \sin(7\pi k + 7\pi h) = (-1)^{7k} \sin(7\pi h) = (-1)^k \sin(7\pi h)$$

$$\sin(\pi(k + h)) = \sin(\pi k + \pi h) = (-1)^k \sin(\pi h)$$

The fraction simplifies as follows:

$$\lim_{h \rightarrow 0} \frac{(-1)^k \sin(7\pi h)}{(-1)^k \sin(\pi h)} = \lim_{h \rightarrow 0} \frac{\sin(7\pi h)}{\sin(\pi h)} = 7$$

The exponent simply evaluates to $\frac{k}{k+1}$ as $x \rightarrow k$. Thus, the sequence L_k is given by:

$$L_k = 7^{\frac{k}{k+1}}$$

Now, we evaluate the outer limit as $k \rightarrow \infty$:

$$\lim_{k \rightarrow \infty} L_k = \lim_{k \rightarrow \infty} 7^{\frac{k}{k+1}}$$

Since $\lim_{k \rightarrow \infty} \frac{k}{k+1} = 1$, we have:

$$\lim_{k \rightarrow \infty} L_k = 7^1 = 7$$

Wednesday 4 March 2026

4.1 Daily Limit

Limit of an Arithmetic Progression Sum

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} (\sin(4) + 3 \sin(4) + 6 \sin(4) + 9 \sin(4) + \cdots + 3n \sin(4))$$

Solution: Observing the series inside the parentheses, excluding the potentially anomalous first term, the terms form an arithmetic progression: $3 \sin(4), 6 \sin(4), 9 \sin(4), \dots, 3n \sin(4)$. We can rewrite the sum as a constant term plus a summation:

$$S_n = \sin(4) + \sum_{k=1}^n 3k \sin(4)$$

Factoring out $3 \sin(4)$ from the summation gives:

$$S_n = \sin(4) + 3 \sin(4) \sum_{k=1}^n k = \sin(4) + 3 \sin(4) \frac{n(n+1)}{2}$$

Now, we substitute this back into the limit:

$$L = \lim_{n \rightarrow \infty} \frac{1}{n^2} \left(\sin(4) + \frac{3n(n+1)}{2} \sin(4) \right)$$

As $n \rightarrow \infty$, the constant term $\frac{\sin(4)}{n^2}$ approaches 0, and we are left with the highest degree terms in the numerator and denominator:

$$L = \lim_{n \rightarrow \infty} \frac{3n^2 + 3n}{2n^2} \sin(4)$$

Dividing numerator and denominator by n^2 :

$$L = \lim_{n \rightarrow \infty} \frac{3 + 3/n}{2} \sin(4) = \frac{3}{2} \sin(4)$$

Thursday 5 March 2026

5.1 Daily Limit

Exponential and Trigonometric Functions Limit

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{e^{\tan(x)} - e^{\sin(x)}}{e^{\tan^{-1}(x)} - e^{\sin^{-1}(x)}}$$

Solution: We can factor out $e^{\sin(x)}$ from the numerator and $e^{\sin^{-1}(x)}$ from the denominator:

$$L = \lim_{x \rightarrow 0} \frac{e^{\sin(x)} \left(e^{\tan(x) - \sin(x)} - 1 \right)}{e^{\sin^{-1}(x)} \left(e^{\tan^{-1}(x) - \sin^{-1}(x)} - 1 \right)}$$

As $x \rightarrow 0$, $e^{\sin(x)} \rightarrow 1$ and $e^{\sin^{-1}(x)} \rightarrow 1$. We use the standard limit $e^u - 1 \sim u$ as $u \rightarrow 0$:

$$L = \lim_{x \rightarrow 0} \frac{\tan(x) - \sin(x)}{\tan^{-1}(x) - \sin^{-1}(x)}$$

We expand the functions using their Maclaurin series up to $O(x^3)$:

$$\begin{aligned}\tan(x) &\approx x + \frac{x^3}{3} \\ \sin(x) &\approx x - \frac{x^3}{6} \\ \tan^{-1}(x) &\approx x - \frac{x^3}{3} \\ \sin^{-1}(x) &\approx x + \frac{x^3}{6}\end{aligned}$$

Substitute these into the numerator and denominator:

$$\text{Numerator} \approx \left(x + \frac{x^3}{3} \right) - \left(x - \frac{x^3}{6} \right) = \frac{x^3}{3} + \frac{x^3}{6} = \frac{x^3}{2}$$

$$\text{Denominator} \approx \left(x - \frac{x^3}{3} \right) - \left(x + \frac{x^3}{6} \right) = -\frac{x^3}{3} - \frac{x^3}{6} = -\frac{x^3}{2}$$

Taking the ratio of the leading terms gives the limit:

$$L = \lim_{x \rightarrow 0} \frac{x^3/2}{-x^3/2} = -1$$

Friday 6 March 2026

6.1 Daily Limit

Rational Limit with Exponential Growth

Evaluate the limit:

$$\lim_{x \rightarrow \infty} \frac{x - e^{\sqrt{x}}}{x + e^{\sqrt{x}}}$$

Solution: As $x \rightarrow \infty$, both x and $e^{\sqrt{x}}$ grow to infinity. However, exponential functions grow significantly faster than polynomial functions. To formalize this, we factor out the dominant term, $e^{\sqrt{x}}$, from both the numerator and the denominator:

$$L = \lim_{x \rightarrow \infty} \frac{e^{\sqrt{x}} \left(\frac{x}{e^{\sqrt{x}}} - 1 \right)}{e^{\sqrt{x}} \left(\frac{x}{e^{\sqrt{x}}} + 1 \right)}$$

Canceling $e^{\sqrt{x}}$ leaves:

$$L = \lim_{x \rightarrow \infty} \frac{\frac{x}{e^{\sqrt{x}}} - 1}{\frac{x}{e^{\sqrt{x}}} + 1}$$

To evaluate $\lim_{x \rightarrow \infty} \frac{x}{e^{\sqrt{x}}}$, we can apply L'Hôpital's Rule or make a substitution $u = \sqrt{x}$, meaning $x = u^2$. As $x \rightarrow \infty$, $u \rightarrow \infty$:

$$\lim_{u \rightarrow \infty} \frac{u^2}{e^u} = 0$$

Since exponentials dominate polynomials, this term vanishes. Substituting this back into our limit yields:

$$L = \frac{0 - 1}{0 + 1} = -1$$

Saturday 7 March 2026

7.1 Daily Limit

Exponential Form Limit with Inverse Tangent

Evaluate the limit:

$$\lim_{x \rightarrow 0} \left(\frac{\tan^{-1}(x)^2}{x^2} \right)^{\left(\frac{\cos(x)^{1/x^2}}{x^2} \right)}$$

Solution: Let the limit be L . This limit is of the form 1^∞ . We can write $L = \exp(\lim_{x \rightarrow 0} E(x) \cdot (B(x) - 1))$, where $B(x)$ is the base and $E(x)$ is the exponent. First, we evaluate the asymptotic behavior of the base $B(x)$:

$$\tan^{-1}(x) \sim x - \frac{x^3}{3} \implies \tan^{-1}(x)^2 \sim \left(x - \frac{x^3}{3} \right)^2 \approx x^2 - \frac{2x^4}{3}$$

$$B(x) = \frac{\tan^{-1}(x)^2}{x^2} \approx \frac{x^2 - 2x^4/3}{x^2} = 1 - \frac{2x^2}{3}$$

Thus, $B(x) - 1 \approx -\frac{2x^2}{3}$. Next, we evaluate the inner term of the exponent $E(x)$:

$$\lim_{x \rightarrow 0} \cos(x)^{1/x^2} = \exp \left(\lim_{x \rightarrow 0} \frac{\ln(\cos x)}{x^2} \right)$$

Using the Maclaurin series $\cos x \approx 1 - x^2/2$, we have $\ln(\cos x) \approx \ln(1 - x^2/2) \approx -x^2/2$.

$$\lim_{x \rightarrow 0} \frac{-x^2/2}{x^2} = -\frac{1}{2} \implies \lim_{x \rightarrow 0} \cos(x)^{1/x^2} = e^{-1/2} = \frac{1}{\sqrt{e}}$$

Now we evaluate the product $E(x) \cdot (B(x) - 1)$:

$$E(x) \cdot (B(x) - 1) = \left(\frac{\cos(x)^{1/x^2}}{x^2} \right) \left(-\frac{2x^2}{3} \right) = -\frac{2}{3} \cos(x)^{1/x^2}$$

Taking the limit as $x \rightarrow 0$:

$$\lim_{x \rightarrow 0} E(x) \cdot (B(x) - 1) = -\frac{2}{3} e^{-1/2} = -\frac{2}{3\sqrt{e}}$$

Exponentiating this result gives the final limit:

$$L = \exp \left(-\frac{2}{3\sqrt{e}} \right)$$

Sunday 8 March 2026

8.1 Daily Limit

Logarithm of Averaged Radicals

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\ln \left(\frac{\sqrt{1+x} + \sqrt{1-x}}{2} \right)}{x^2}$$

Solution: We utilize the binomial series expansion for $\sqrt{1+u} = (1+u)^{1/2}$ up to $O(x^2)$:

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + O(x^3)$$

$$\sqrt{1-x} = 1 - \frac{x}{2} - \frac{x^2}{8} + O(x^3)$$

Adding the two expressions cancels the linear terms:

$$\sqrt{1+x} + \sqrt{1-x} = 2 - \frac{x^2}{4} + O(x^4)$$

Dividing by 2 gives the argument of the natural logarithm:

$$\frac{\sqrt{1+x} + \sqrt{1-x}}{2} = 1 - \frac{x^2}{8} + O(x^4)$$

Substitute this back into the limit:

$$L = \lim_{x \rightarrow 0} \frac{\ln \left(1 - \frac{x^2}{8} + O(x^4) \right)}{x^2}$$

Using the Maclaurin series expansion for $\ln(1+u) \approx u$ as $u \rightarrow 0$, where $u = -\frac{x^2}{8}$:

$$\ln \left(1 - \frac{x^2}{8} \right) \approx -\frac{x^2}{8}$$

Therefore, the limit is:

$$L = \lim_{x \rightarrow 0} \frac{-x^2/8}{x^2} = -\frac{1}{8}$$

Monday 9 March 2026

9.1 Daily Limit

Limit of an n -th Root of a Sine Product

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \sqrt[n]{\prod_{k=1}^{n-1} \sin\left(\frac{k\pi}{2n}\right)}$$

Solution: Let $P_n = \prod_{k=1}^{n-1} \sin\left(\frac{k\pi}{2n}\right)$. We can evaluate this product using properties of roots of unity, which give the well-known identity for the full period:

$$\prod_{k=1}^{2n-1} \sin\left(\frac{k\pi}{2n}\right) = \frac{2n}{2^{2n-1}}$$

By symmetry, $\sin\left(\frac{(2n-k)\pi}{2n}\right) = \sin\left(\pi - \frac{k\pi}{2n}\right) = \sin\left(\frac{k\pi}{2n}\right)$. The central term at $k = n$ is $\sin\left(\frac{\pi}{2}\right) = 1$. Thus, the product splits into two identical halves:

$$P_n \cdot 1 \cdot P_n = \frac{2n}{2^{2n-1}} \implies P_n^2 = \frac{n}{2^{2n-2}} \implies P_n = \frac{\sqrt{n}}{2^{n-1}}$$

Now, we substitute P_n into the limit :

$$L = \lim_{n \rightarrow \infty} (P_n)^{1/n} = \lim_{n \rightarrow \infty} \left(\frac{n^{1/2}}{2^{n-1}}\right)^{1/n}$$

Distributing the $1/n$ power:

$$L = \lim_{n \rightarrow \infty} \frac{n^{1/2n}}{2^{1-1/n}}$$

As $n \rightarrow \infty$, $n^{1/n} \rightarrow 1$, so the numerator approaches $1^{1/2} = 1$. The exponent in the denominator approaches 1.

$$L = \frac{1}{2^1} = \frac{1}{2}$$

Alternatively, applying Riemann sums to $\frac{1}{n} \ln(P_n)$ converts it to the integral $\int_0^1 \ln\left(\sin\left(\frac{\pi x}{2}\right)\right) dx = -\ln 2$, leading to $L = e^{-\ln 2} = \frac{1}{2}$.

Tuesday 10 March 2026

10.1 Daily Limit

Infinite Product with Alternating Exponents

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \prod_{k=1}^n \left(1 + \frac{1}{k}\right)^{(-1)^k}$$

Solution: Let P_n denote the n -th partial product. We analyze the terms by grouping them in pairs of odd and even indices. For an odd index $k = 2m - 1$:

$$\left(1 + \frac{1}{2m-1}\right)^{-1} = \left(\frac{2m}{2m-1}\right)^{-1} = \frac{2m-1}{2m}$$

For an even index $k = 2m$:

$$\left(1 + \frac{1}{2m}\right)^1 = \frac{2m+1}{2m}$$

Multiplying the $(2m-1)$ -th and $(2m)$ -th terms together yields a single combined term for each pair m :

$$\left(\frac{2m-1}{2m}\right) \left(\frac{2m+1}{2m}\right) = \frac{4m^2-1}{4m^2}$$

The partial product for an even number of terms, $n = 2N$, can be written as:

$$P_{2N} = \prod_{m=1}^N \frac{4m^2-1}{4m^2}$$

We recognize the reciprocal of this product from the Wallis product formula for π , which states:

$$\frac{\pi}{2} = \prod_{m=1}^{\infty} \frac{4m^2}{4m^2-1}$$

Therefore, taking the limit as $N \rightarrow \infty$, our product is the reciprocal of the Wallis product:

$$\lim_{N \rightarrow \infty} P_{2N} = \left(\frac{\pi}{2}\right)^{-1} = \frac{2}{\pi}$$

(The odd partial products P_{2N-1} converge to the same limit since the extra factor $\frac{2N+1}{2N+2} \rightarrow 1$).

$$L = \frac{2}{\pi}$$

Wednesday 11 March 2026

11.1 Daily Limit

Limit Involving Zeta, Error Function, and an Infinite Product

Evaluate the limit $\lim_{x \rightarrow \infty} f(x)$, where $f(x)$ is defined as:

$$f(x) = \frac{\zeta(x^{-x}) \operatorname{erf}\left(\frac{x\sqrt{\ln(x)}}{\sqrt{x^2+1}}\right)}{\prod_{k=1}^x \left(1 + \frac{1}{k}\right) e^{-\frac{1}{k}}}$$

Solution: We can evaluate this limit by analyzing the three main structural components of $f(x)$ independently as $x \rightarrow \infty$.

1. The Riemann Zeta Function Component: Consider the argument of the zeta function, x^{-x} . As $x \rightarrow \infty$, x^{-x} converges to 0. We rely on the known value of the Riemann zeta function evaluated at zero:

$$\lim_{x \rightarrow \infty} \zeta(x^{-x}) = \zeta(0) = -\frac{1}{2}$$

2. The Error Function Component: We evaluate the limit of the argument inside the error function:

$$\lim_{x \rightarrow \infty} \frac{x\sqrt{\ln(x)}}{\sqrt{x^2+1}} = \lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2(1+\frac{1}{x^2})}} \sqrt{\ln(x)} = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{1+\frac{1}{x^2}}} \sqrt{\ln(x)}$$

As $x \rightarrow \infty$, the fraction approaches 1 and $\sqrt{\ln(x)} \rightarrow \infty$, so the entire argument goes to ∞ . The error function $\operatorname{erf}(u)$ has a horizontal asymptote at 1 as $u \rightarrow \infty$. Thus:

$$\lim_{x \rightarrow \infty} \operatorname{erf}\left(\frac{x\sqrt{\ln(x)}}{\sqrt{x^2+1}}\right) = 1$$

3. The Product Component (Denominator): We interpret the product limit assuming integer upper bounds $n \rightarrow \infty$. Let P_n be the partial product:

$$P_n = \prod_{k=1}^n \left(1 + \frac{1}{k}\right) e^{-1/k} = \left(\prod_{k=1}^n \frac{k+1}{k}\right) \exp\left(-\sum_{k=1}^n \frac{1}{k}\right)$$

The first term is a telescoping product:

$$\prod_{k=1}^n \frac{k+1}{k} = \frac{2}{1} \cdot \frac{3}{2} \cdot \frac{4}{3} \cdots \frac{n+1}{n} = n+1$$

The summation in the exponent is exactly the n -th harmonic number H_n . We use the asymptotic expansion $H_n = \ln(n) + \gamma + O\left(\frac{1}{n}\right)$, where γ is the Euler-Mascheroni constant. Substituting these into P_n :

$$P_n = (n+1) e^{-(\ln(n) + \gamma + O(\frac{1}{n}))} = (n+1) \cdot \frac{1}{n} \cdot e^{-\gamma} \cdot e^{-O(\frac{1}{n})}$$

$$P_n = \left(1 + \frac{1}{n}\right) e^{-\gamma} e^{-O(\frac{1}{n})}$$

Taking the limit as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} P_n = (1) \cdot e^{-\gamma} \cdot (1) = e^{-\gamma}$$

Final Evaluation: We can now combine the limits of the individual components back into the original fraction:

$$\lim_{x \rightarrow \infty} f(x) = \frac{\left(-\frac{1}{2}\right)(1)}{e^{-\gamma}} = -\frac{e^{\gamma}}{2}$$